

Sequential Models for Swahili News Topic Classification: Evidence from Five Approaches

Formative Assignment 2, Research-Informed Sequential Models for NLP and Language Technologies

Gentil Tonny Christian Iradukunda, Dan Paul Dushime, Rene Pierre Ntabana, Thierry Alain Tresor Ibyishaka

Source code: github.com/irachrist1/swahili-news-classification · Demo video: [\[add link\]](#) · Contribution tracker: [\[add link\]](#)

1. Introduction

Swahili is spoken by an estimated 100 to 150 million people across East Africa and is a national language of Tanzania, where it is also the language of instruction in primary schools. A large share of Tanzanian news is published in Swahili, yet most tools that route, tag and recommend news are built for English. Automatic topic classification is a basic building block for such tools: it lets a newsroom file incoming stories, lets an archive be searched by subject, and lets a reader follow only health or economy coverage. For low-resource languages it is also a practical test of whether modern NLP methods transfer beyond English (Joshi et al., 2020).

This study investigates how well sequential models classify Swahili news articles into six topics: *uchumi* (economy), *kitaifa* (national), *michezo* (sports), *kimataifa* (international), *burudani* (entertainment) and *afya* (health). The guiding research question is:

How effectively can sequential modelling approaches classify Swahili news articles, and what evidence explains the strengths and limitations of each approach?

We compare two bag-of-words baselines, TF-IDF with Naive Bayes and TF-IDF with Logistic Regression, against three neural architectures that model word order in different ways: a bidirectional LSTM with attention (recurrence), a TextCNN (local convolution) and a fine-tuned AfriBERTa transformer (self-attention with African-language pretraining). Each model family was developed through a tracked sequence of experiments in which each run changes one factor and is kept only if it improves validation macro-F1. The objective is not only to find the best score, but to explain *why* the models differ using the data, the experiments and prior research.

2. Related Work

Swahili and African-language text classification. The dataset used here was released by Davis (2020) through the AI4D language dataset programme and used for the Zindi *AI4D Swahili News Classification Challenge* (Zindi, 2021), which scored submissions by multi-class log loss. Adelani et al. (2023) later built MasakhaNEWS, a news-topic benchmark for 16 African languages including Swahili. They compared classical machine-learning baselines with fine-tuned language models and found that fine-tuned African-centric models such as AfroXLMR (Alabi et al., 2022) gave the strongest fully supervised results, and that few-shot methods recovered most of that performance with only 10 labelled examples per class. SwahBERT (Martin et al., 2022) showed that a Swahili-specific BERT improves several downstream Swahili tasks over multilingual BERT. These results motivated including both strong classical baselines and a pretrained African-language transformer, so that we could measure how much pretraining adds on this dataset.

Bag-of-words baselines. Wang and Manning (2012) showed that Naive Bayes and linear models over word n-grams are hard to beat for topic classification, because topic is carried mostly by *which*

words appear rather than their order. Joulin et al. (2017) reached similar conclusions with fastText. Rennie et al. (2003) proposed Complement Naive Bayes to correct the standard model’s bias towards large classes. Given the strong class imbalance we found (Section 3), we tested it directly.

Recurrent models. Long short-term memory networks (Hochreiter & Schmidhuber, 1997) keep a memory of earlier words through gated recurrence, and bidirectional processing (Schuster & Paliwal, 1997) lets each word state see both left and right context. For long documents, a single final hidden state must compress the whole article, so attention pooling (Bahdanau et al., 2015; Yang et al., 2016) instead learns a weighted average over all word states. Our articles have a median length of 270 words, so this was a design question we tested rather than assumed.

Convolutional models. Kim (2014) showed that a single layer of 1D convolutions with several filter widths, followed by max-pooling over time, performs well on sentence classification. The convolutions act as learned n-gram detectors that fire wherever a key phrase appears (Zhang et al., 2015). This is a different sequential inductive bias from the LSTM: local order within a short window, no long-range memory, and much faster training.

Transformers and pretraining. Self-attention (Vaswani et al., 2017) lets every token attend to every other token in the window, and BERT-style pretraining (Devlin et al., 2019) transfers knowledge learned from large unlabelled corpora. AfriBERTa (Ogueji et al., 2021) was pretrained on under 1 GB of text in 11 African languages, including Swahili, and was competitive with much larger multilingual models. We chose its small version (4 layers, 83 million parameters) because it fits our compute budget and its pretraining matches the target language. Its window is limited to 512 subword tokens, so we followed Sun et al. (2019), who compared truncation strategies for long documents and found that keeping the head and the tail of a document worked best.

Word embeddings. Grave et al. (2018) released fastText vectors for 157 languages, including Swahili, trained on Common Crawl and Wikipedia with subword information (Bojanowski et al., 2017). We tested them as initialisation for the BiLSTM and TextCNN because more than half of our training vocabulary appears only once.

Evaluation. Sokolova and Lapalme (2009) show that accuracy hides poor performance on small classes, and Opitz and Burst (2019) recommend macro-F1, the unweighted mean of per-class F1, when all classes matter equally. Log loss rewards well-calibrated probabilities, and modern neural networks are often over-confident (Guo et al., 2017). For comparing two classifiers on the same test set, Dietterich (1998) recommends McNemar’s test, and Dror et al. (2018) argue that NLP results should be reported with significance tests rather than single scores.

Table 1. How the literature and the data analysis shaped each approach.

Approach	Sequential inductive bias	Why it was included
TF-IDF + Naive Bayes	None (bag of words, bigrams give local order)	Classic strong topic baseline (Wang & Manning, 2012); Complement NB tested for imbalance (Rennie et al., 2003)
TF-IDF + Logistic Regression	None (word and character n-grams)	Strong discriminative baseline; character n-grams share evidence across Swahili's agglutinative word forms
BiLSTM + attention	Full left-to-right and right-to-left recurrence	Captures long-range context; attention addresses long articles (Yang et al., 2016)
TextCNN	Local windows of 2 to 5 words, position-free max-pooling	Topic cues are key phrases; fast and robust (Kim, 2014)
AfriBERTa-small	Self-attention over 512 subword tokens, pretrained	African-language pretraining (Ogueji et al., 2021); best results in MasakhaNEWS came from such models (Adelani et al., 2023)

3. Dataset and Exploratory Analysis

Source. The Swahili News Classification dataset (Davis, 2020; CC BY 4.0) contains articles collected from Tanzanian Swahili news websites, each labelled with one of six topics. The Zindi challenge keeps its test labels hidden, so we used the labelled release on Zenodo (22,207 training and 7,338 test articles), downloaded from its Hugging Face mirror. This gives a large, labelled, held-out test set and needs no login, which keeps the pipeline reproducible on Colab.

Cleaning the splits. We removed 3 articles with fewer than 20 words (one was empty, one was just ['.']) and 34 test articles that also appear word-for-word in the training set, which would otherwise leak answers into the test score. We then held out a stratified 10% of the training set for validation. The final splits are 19,984 training, 2,221 validation and 7,303 test articles. All model selection used validation only; the test set was used once per model family, for the final configuration.

Class imbalance. *Kitaifa* (national news) makes up 45.7% of training articles, while *afya* (health) makes up 3.5%, a ratio of 13 to 1 (Figure 1). A model that always answers *kitaifa* already reaches 45.7% accuracy. This finding drove three decisions: macro-F1 as the main metric, class-weighted loss as an experiment for every model family, and Complement Naive Bayes as an experiment for the Naive Bayes baseline.

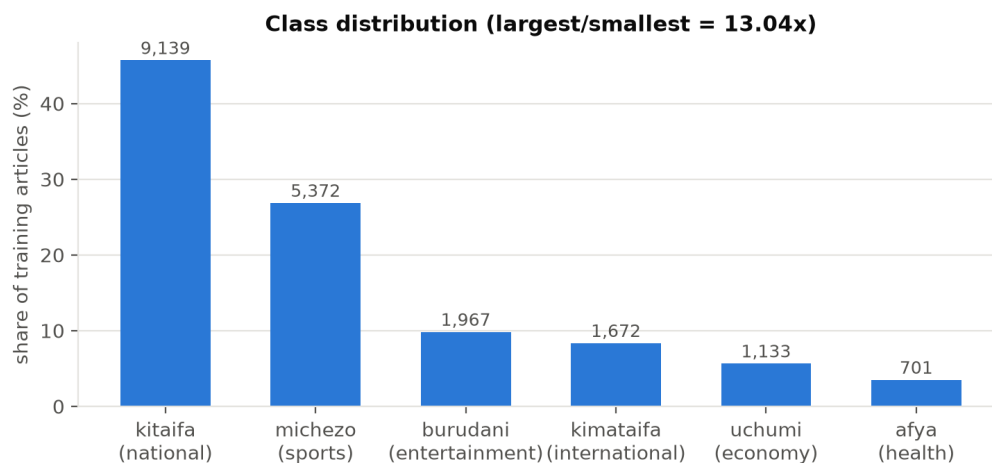


Figure 1. Class distribution of the training set.

Sequence length. Articles are long for sequence models: the median is 270 words, the 90th percentile is 592, and the longest has 3,840 words (Figure 2). 54% of articles are longer than 256

words and 14% are longer than 512. AfriBERTa’s tokenizer produces 1.41 subword tokens per word, so the median article is 378 tokens and 29.5% of articles exceed the transformer’s 512-token window. Length also differs by class: entertainment articles are the shortest (median 136 words) and health articles the longest (median 348). Truncation length was therefore a key experimental variable for all three neural models.

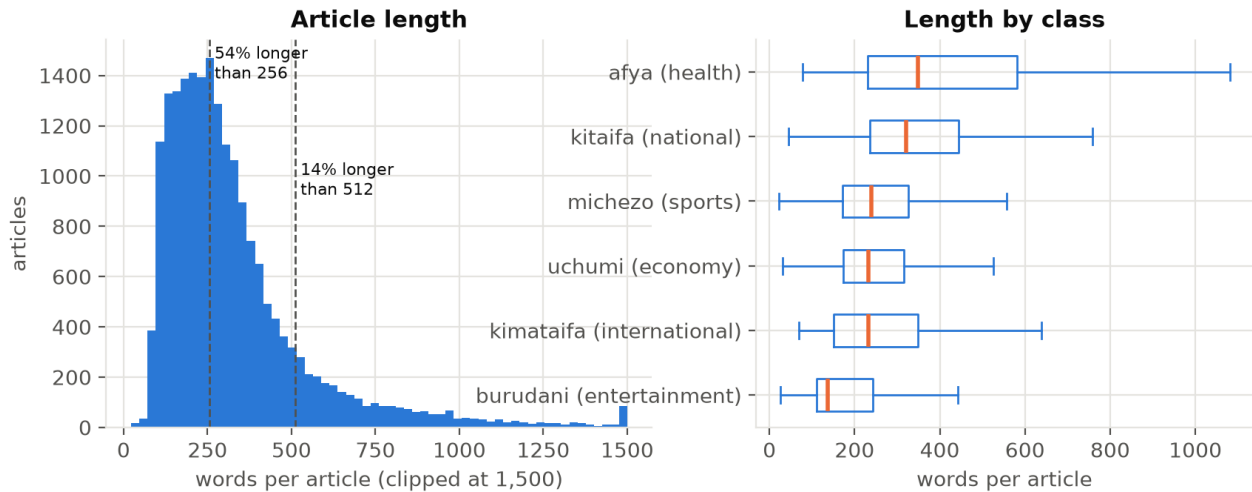


Figure 2. Article length overall and by class.

Where the signal sits. To decide how to truncate, we trained a TF-IDF Logistic Regression on only the first k or only the last k words of each article (Figure 3). The first 16 words alone reach 0.785 validation macro-F1, and the first 128 words reach 0.819, almost as much as 512 words (0.825). The last 16 words reach only 0.584. News follows the “inverted pyramid”: the headline-like opening sentence names the topic, and many articles open with a byline such as “*Na NORA DAMIAN – DAR ES SALAAM*” (10.3% of articles). This suggested that head truncation is safe, and that the extra text mostly helps the hardest cases.

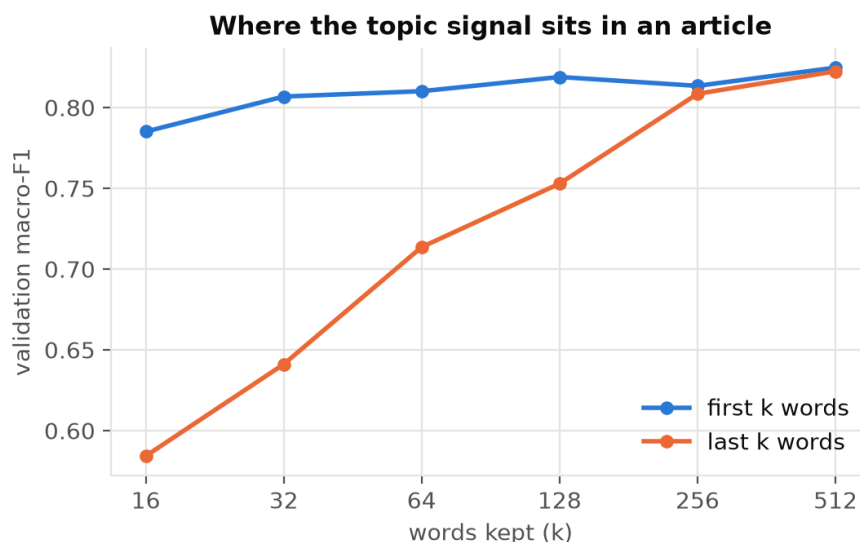


Figure 3. Validation macro-F1 of a linear model that sees only the first or last k words.

Vocabulary and rare words. The training set has 6.7 million tokens and 189,739 distinct words after cleaning. The distribution is strongly Zipfian: 52.5% of distinct words appear only once, a sign of Swahili’s rich agglutinative morphology (*wachezaji*, *mchezaji*, *kuchezea* all share the root *-cheza*, “play”), names, and spelling variation. Keeping words seen at least twice gives a 90,209-word

vocabulary that covers 98.5% of tokens. The neural models use the 50,000 most frequent words, and the validation out-of-vocabulary rate is 2.6%. fastText Swahili vectors cover 77.8% of this vocabulary. The rare-word tail motivated character n-grams for Logistic Regression, pretrained embeddings for the BiLSTM and TextCNN, and subword tokenization for AfriBERTa.

Noise from scraping. Two systematic problems appeared on close reading. First, 47% of articles contain sentences glued together without a space (“*mwaka huu.Misafara hiyo*”), on average 3.1 times per article, which creates false tokens. Second, the “fi” and “fl” ligatures were broken during extraction, splitting words such as *ofisi* (office) into “*Ofi si*” and *kufikia* (to reach) into “*kufi kia*”. We found 2,412 such candidate splits. Naively joining every “fi” split would break real phrases such as *safi na* (“clean and”), so we repaired a split only if the joined word appears in the corpus at least twice as often as the split form. This data-driven rule learned 127 repairs from the training set and fixed 653 occurrences while leaving *safi na* untouched.

Code-switching. 12.1% of articles contain at least one English function word, but only 2.3% have more than 1% English words. Code-switching is concentrated in entertainment (10.3% of articles above the threshold), where artist names, song titles and English phrases are common, compared with under 1% for national and health news.

Lexical separability and shortcuts. The words most associated with each class (chi-squared on TF-IDF, keeping only positive associations) are shown in Table 2. Sports and entertainment have very distinctive vocabularies (club names *Yanga* and *Simba*; *msanii*, artist), which predicts high scores for these classes. Economy and national news share vocabulary about government projects and business, and some cues are named entities tied to one period: *Magufuli* (Tanzania’s president from 2015 to 2021), *Trump*, *Mugabe* and the company *Airtel*. Models that lean on such names may not generalise to news from a different period, a point we return to in Section 6.

Table 2. Most class-associated words in the training set.

uchumi (economy)	kitaifa (national)	michezo (sports)	kimataifa (international)	burudani (entertainment)	afya (health)
airtel, wateja, benki, bidhaa, biashara, uwekezaji	wananchi, serikali, magufuli, wilaya, dk, ujenzi	timu, mchezo, yanga, ligi, simba, wachezaji	trump, korea, iran, marekani, urusi, kenyatta	muziki, msanii, wasanii, wimbo, filamu, diamond	anasema, saratani, damu, wagonjwa, ugonjwa, afya

4. Methodology

4.1 Preprocessing and sequence representation

All models share the same splits and the same two cleaning steps described above (splitting glued sentences, learned ligature repair), both fitted on the training set only. After that, the representation depends on the model:

- **TF-IDF baselines:** lowercased words, numbers replaced by a <num> token, punctuation removed. Features are sublinear TF-IDF over word unigrams and bigrams (at most 300,000 features, minimum document frequency 2) and, for Logistic Regression, character 2 to 5-grams within word boundaries.
- **BiLSTM and TextCNN:** the same cleaned text, mapped to ids over a 50,000-word vocabulary (unknown words map to <unk>), truncated to the first 256 or 512 words. To keep training fast without packed sequences, batches group articles of similar length and are padded to a multiple of 32.

- **AfriBERTa:** light cleaning only (casing and punctuation are kept, because its SentencePiece tokenizer was trained on raw text), truncated to 256 or 512 subword tokens, using either the head or the head plus tail (first 128 and last 382 tokens) of the article. Batches are padded to the full window.

4.2 Models

1. **Naive Bayes:** Multinomial or Complement NB over TF-IDF features, with the smoothing parameter α tuned on validation.
2. **Logistic Regression:** multinomial logistic regression (L2, lbfgs) with the inverse regularisation strength C tuned on validation.
3. **BiLSTM + attention:** 300-dimensional embeddings, a one-layer bidirectional LSTM with 128 units per direction, and either the final hidden states or additive attention pooling over all states, followed by dropout (0.3) and a linear layer (15.4 million parameters, 15.0 million of them in the embedding).
4. **TextCNN:** 300-dimensional embeddings, parallel 1D convolutions (100 filters each of widths 3, 4 and 5, or 200 filters of widths 2 to 5), ReLU, max-pooling over time, dropout (0.5) and a linear layer (15.4 million parameters).
5. **AfriBERTa-small:** the pretrained `castorini/afriberta_small` encoder (4 layers, hidden size 768, 83 million parameters) with a classification head on the first token, fully fine-tuned.

Neural models were trained with AdamW (learning rate $1e-3$ for BiLSTM and TextCNN, $5e-5$ with 10% linear warm-up and linear decay for AfriBERTa), gradient clipping at 1.0 and cross-entropy loss. BiLSTM and TextCNN trained for up to 8 epochs with early stopping (patience 2) on validation macro-F1; AfriBERTa trained for 3 epochs and kept the best epoch. The best checkpoint by validation macro-F1 was restored before evaluation.

4.3 Experimental design

Each model family followed a written plan (in `src/baselines.py`, `src/train_neural.py` and `src/train_transformer.py`). Every experiment states the single change it makes and the hypothesis behind it, and it is logged to `results/experiment_log.csv`. For the neural models the plan is greedy: each step changes one factor relative to the best configuration so far, and the change is kept only if it improves validation macro-F1. The factors cover the dimensions the assignment asks for:

- **Preprocessing:** raw versus cleaned text, and stopword removal (Naive Bayes).
- **Representation:** word unigrams versus bigrams, character n-grams (Logistic Regression), random versus fastText embeddings (BiLSTM, TextCNN), and 256 versus 512 words or tokens with head or head+tail truncation (all neural models).
- **Architecture:** final hidden state versus attention pooling (BiLSTM), filter widths and count (TextCNN).
- **Class balancing:** Complement NB, balanced class weights (Logistic Regression) and class-weighted cross-entropy (BiLSTM, TextCNN).
- **Hyperparameters:** α for Naive Bayes and C for Logistic Regression, by grid search.

Class-weighted loss was not tested for AfriBERTa. Each 512-token AfriBERTa run takes well over an hour on our hardware (Section 5.1), the class-weight hypothesis had already been tested on three other model families, and we preferred to spend that budget on a second random seed.

Robustness. The final BiLSTM and TextCNN configurations were retrained with three seeds (42, 7, 123) and AfriBERTa with two (42, 7); we report the mean and standard deviation of test macro-F1. The

baselines are deterministic. For each final model we also report a 95% bootstrap confidence interval for test macro-F1 (1,000 resamples) and run exact McNemar tests between every pair of models.

4.4 Evaluation metrics

- **Macro-F1 (primary):** the unweighted mean of the six per-class F1 scores. Because it gives *afya* (3.5% of articles) the same weight as *kitaifa* (45.7%), it rewards models that work for every topic, which is what a newsroom tagging system needs (Opitz & Burst, 2019; Sokolova & Lapalme, 2009).
- **Accuracy and weighted F1:** reported for comparison, but both are dominated by the two large classes.
- **Log loss:** the Zindi challenge’s official metric. It measures the quality of the predicted probabilities, not just the top label, and heavily penalises confident mistakes. We also report expected calibration error and reliability curves (Guo et al., 2017).
- **Macro ROC-AUC (one-vs-rest):** measures ranking quality independently of the decision threshold.
- **Per-class F1 and confusion matrices:** show *which* topics each model confuses.

The limitation of this strategy is that macro-F1 treats all errors as equally costly and is sensitive to a handful of examples in the smallest class: one extra correct *afya* article moves macro-F1 about as much as fifteen extra correct *kitaifa* articles. We therefore report confidence intervals and per-class scores alongside it.

5. Results and Discussion

5.1 Experiment progression

Figure 4 shows every tracked experiment in the order it was run. Each line starts from a simple configuration, and the dashed line shows the best validation macro-F1 so far.

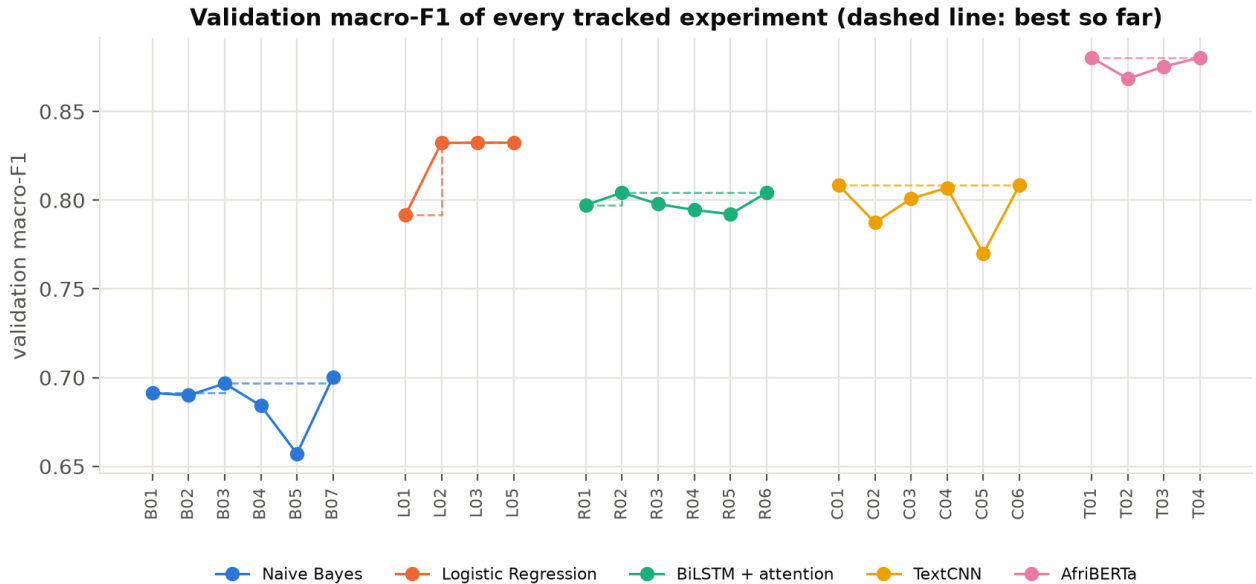


Figure 4. Validation macro-F1 of every experiment. B = Naive Bayes, L = Logistic Regression, R = BiLSTM, C = TextCNN, T = AfriBERTa. The last point of each family is the selected configuration.

Baselines. Cleaning the text barely changed Naive Bayes (0.691 raw, 0.690 cleaned), which suggests that glued sentences and broken ligatures affect too few topic words to matter for a bag-of-words model. Removing stopwords helped slightly (0.697), bigrams did not (0.684), and Complement NB

only helped once its smoothing was tuned ($\alpha = 0.03$, 0.700). For Logistic Regression, balanced class weights gave the largest single gain of the whole study, from 0.791 to 0.832, confirming that imbalance, not representation, was the main obstacle. Character n-grams added almost nothing (0.832), and $C = 10$ was best.

BiLSTM. Attention pooling improved on the final hidden state (0.797 $\rightarrow$ 0.804) and was kept. The three later hypotheses all failed: 512 words (0.798), fastText initialisation (0.794) and class weights (0.792). **TextCNN.** Nothing beat the first configuration (0.808): 512 words gave 0.787, fastText 0.801, class weights 0.807 and wider filters 0.770.

The seed runs put these small differences in perspective. The final BiLSTM’s validation macro-F1 ranged from 0.804 to 0.822 across three seeds, and the TextCNN’s from 0.785 to 0.808. Most single-seed steps in the plan moved by less than that, so for these two models we can only conclude that none of the changes gave a *reliable* gain. This is itself a finding: on this dataset, longer inputs, pretrained word vectors and class weighting do not rescue sequence models trained from scratch.

AfriBERTa. Starting at 0.880, AfriBERTa was already better than every other model after its first experiment. Reading 512 tokens instead of 256 (To2) did not improve validation macro-F1 (0.868 against 0.880), and head+tail truncation (To3) gave 0.875, so, unlike Sun et al. (2019), keeping the end of the article did not help. This agrees with our positional analysis: the last words of an article carry much less topic signal than the first. The final configuration uses 256 tokens with head truncation. One run of it took 42 minutes on our laptop GPU and each 512-token run about 68 minutes, compared with 9 minutes for the BiLSTM and 5 minutes for the TextCNN.

The full experiment log, with the hypothesis behind every run, is in Appendix A and `results/experiment_log.csv`.

5.2 Model comparison

Table 3. Test-set results of the five selected configurations ($n = 7,303$). The 95% interval is a bootstrap interval for macro-F1; seed mean $\pm$ SD is over three seeds (two for AfriBERTa).

Model	Val macro-F1	Test macro-F1	95% CI	Seeds mean $\pm$ SD	Accuracy	Weighted F1	Log loss	Macro AUC	ECE	Params
Naive Bayes	0.700	0.689	0.674–0.703	–	0.868	0.841	0.623	0.935	0.080	–
Logistic Regression	0.832	0.817	0.805–0.829	–	0.889	0.890	0.333	0.979	0.018	–
BiLSTM + attention	0.804	0.795	0.782–0.809	0.800 $\pm$ 0.004	0.882	0.881	0.379	0.975	0.043	15.4M
TextCNN	0.808	0.791	0.777–0.806	0.781 $\pm$ 0.009	0.882	0.879	0.381	0.975	0.034	15.4M
AfriBERTa	0.880	0.868	0.857–0.879	0.870 $\pm$ 0.001	0.925	0.924	0.289	0.987	0.049	83.1M

AfriBERTa is the best model on every metric except calibration (Table 3). It reaches 0.868 test macro-F1 against 0.817 for Logistic Regression, the strongest non-pretrained model, and the exact McNemar test shows the difference in errors is very unlikely to be chance (Table 4). The two sequential models trained from scratch did **not** beat the tuned linear baseline: the BiLSTM reached 0.795 (0.800 $\pm$ 0.004 over seeds) and the TextCNN 0.791 (0.781 $\pm$ 0.009), and McNemar tests show Logistic Regression makes significantly fewer errors than both ($p = 0.031$ and 0.026). Naive Bayes is clearly weakest (0.689).

Table 4. Exact McNemar tests between every pair of models on the test set, counting only the articles where exactly one of the two is correct.

Comparison	Only first correct	Only second correct	p-value
AfriBERTa vs Logistic Regression	446	187	2.4e-25
AfriBERTa vs BiLSTM + attention	489	178	2e-34
AfriBERTa vs TextCNN	488	177	1.5e-34
AfriBERTa vs Naive Bayes	609	190	6.3e-52
Logistic Regression vs BiLSTM + attention	305	253	0.031
Logistic Regression vs TextCNN	288	236	0.026
Logistic Regression vs Naive Bayes	476	316	1.4e-08
BiLSTM + attention vs TextCNN	256	256	1
BiLSTM + attention vs Naive Bayes	453	345	0.00015
TextCNN vs Naive Bayes	386	278	3.2e-05

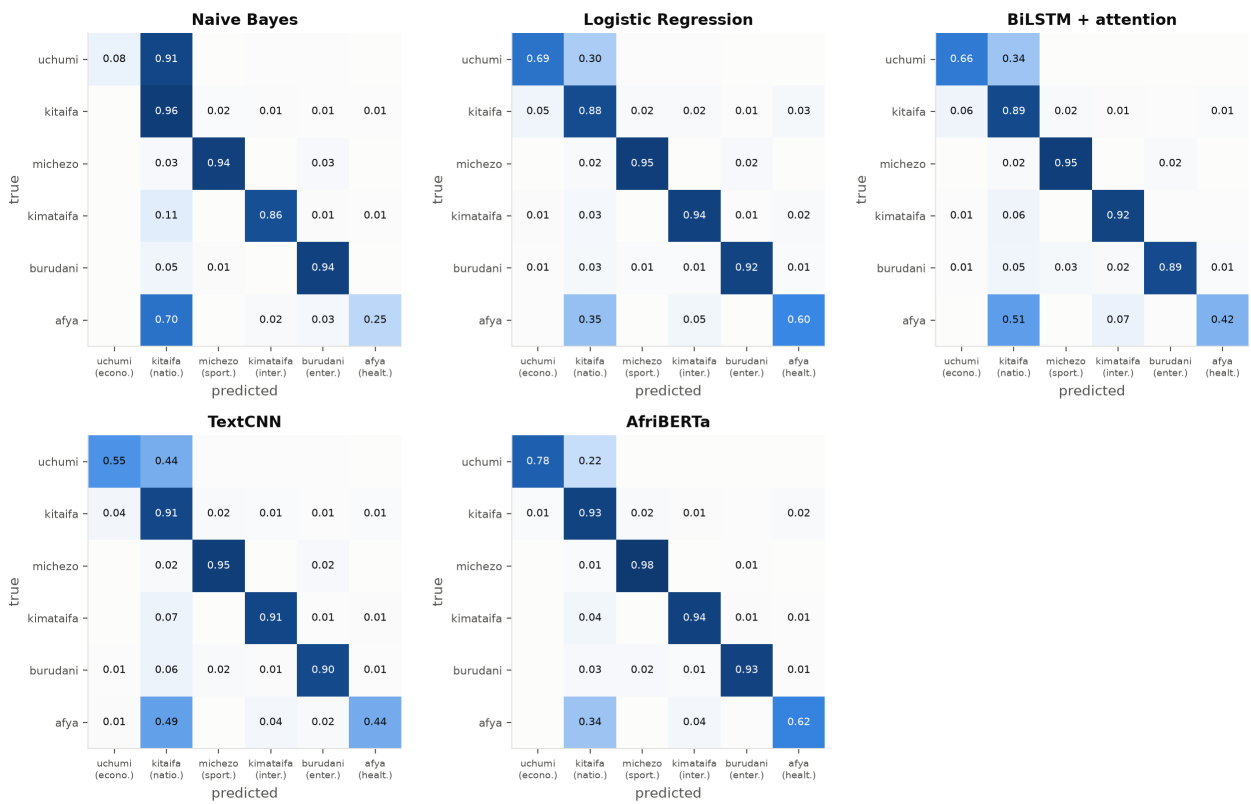


Figure 5. Row-normalised test confusion matrices.

Per-class results. Accuracy hides where the models differ. Sports, international and entertainment news are easy for every model (F1 around 0.90 to 0.96), because their vocabularies are distinctive (Table 2). The differences are concentrated in the two small, overlapping classes, economy and health (Table 5, Figure 6).

Table 5. Test F1 per class.

Model	uchumi	kitaifa	michezo	kimataifa	burudani	afya
Naive Bayes	0.14	0.88	0.96	0.89	0.91	0.36
Logistic Regression	0.67	0.90	0.96	0.90	0.91	0.56
BiLSTM + attention	0.63	0.89	0.95	0.90	0.90	0.50
TextCNN	0.59	0.89	0.96	0.90	0.89	0.51
AfriBERTa	0.83	0.93	0.97	0.93	0.94	0.62

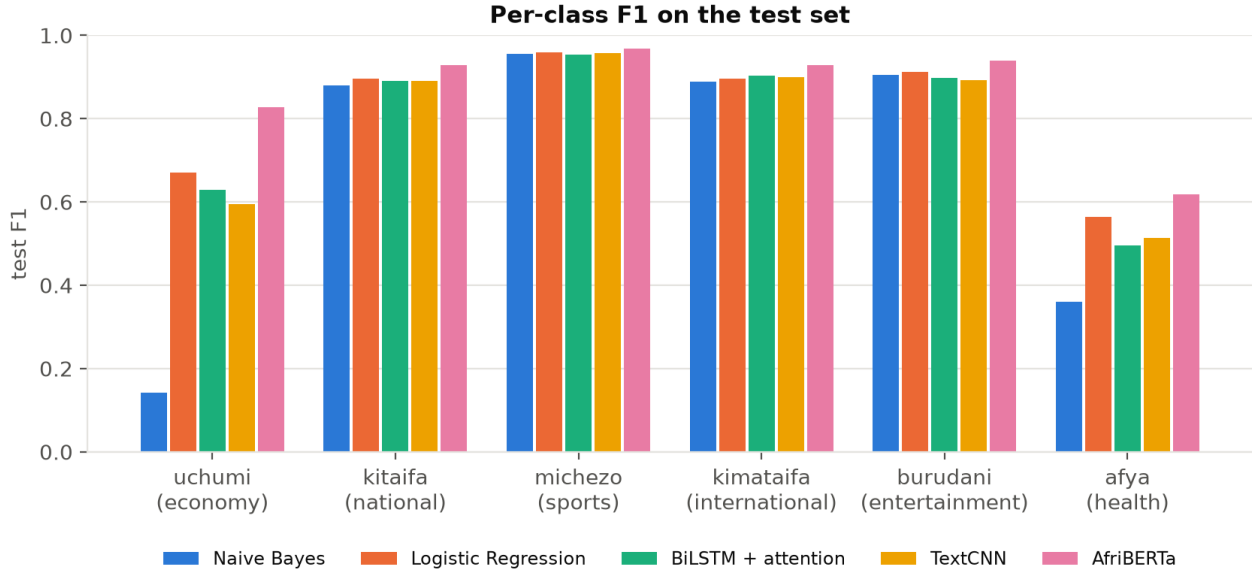


Figure 6. Per-class test F1.

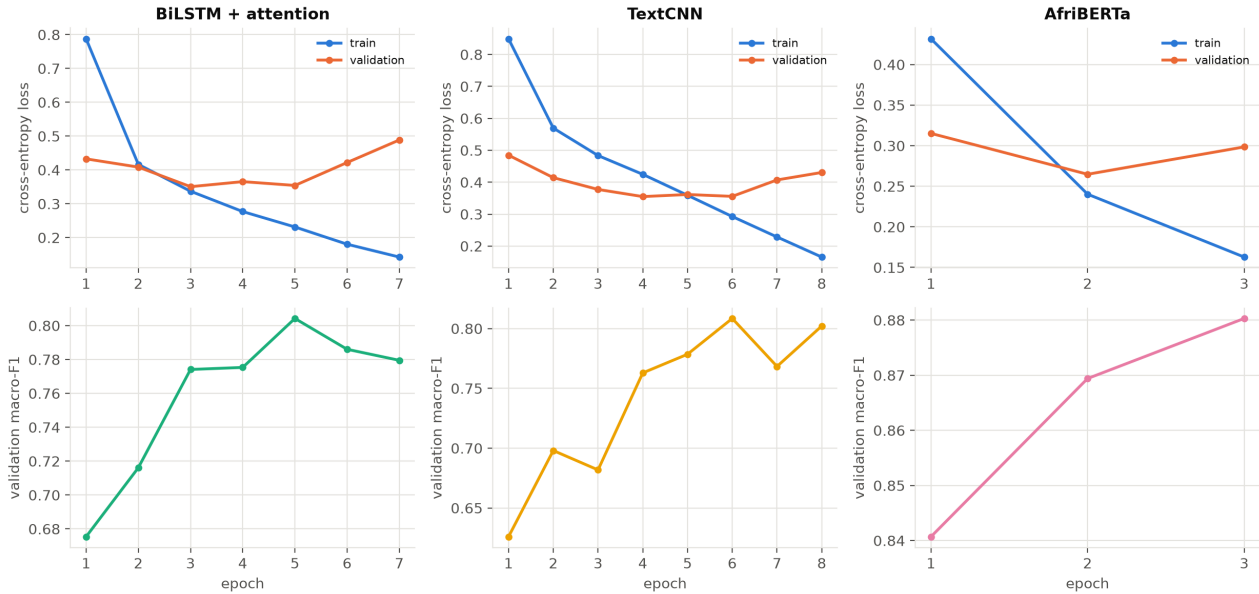
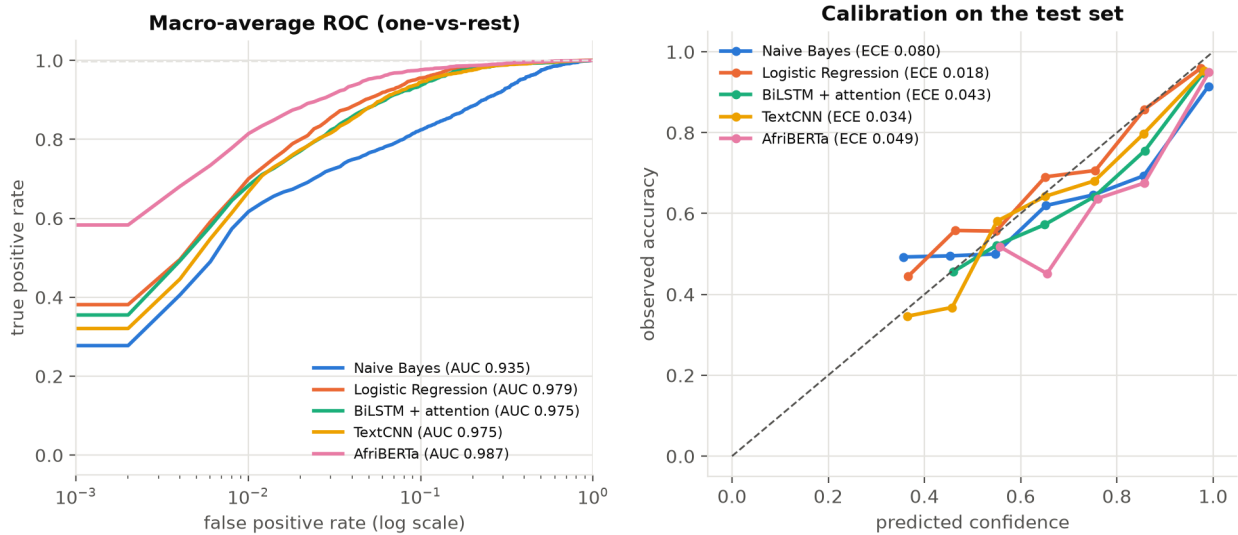


Figure 7. Learning curves of the final neural models (seed 42). Top: training and validation loss. Bottom: validation macro-F1.

Learning curves (Figure 7). Both models trained from scratch fit the training set quickly: training loss keeps falling while validation loss bottoms out after three to six epochs and then rises, which is why early stopping on validation macro-F1 was essential. AfriBERTa’s validation loss is lower than theirs from the first epoch, the benefit of starting from pretrained representations rather than random embeddings for 50,000 words. In its third epoch, its validation loss rises while macro-F1 still improves: the model gets more articles right but becomes over-confident on the ones it gets wrong.



Ranking and calibration (Figure 8). All models except Naive Bayes rank articles well (macro AUC ≥ 0.975), so their differences are mostly in the hard decisions near class boundaries. Log loss, the Zindi metric, orders the models the same way as macro-F1. Naive Bayes is badly calibrated (ECE 0.080): its independence assumption double-counts correlated words and pushes probabilities towards 0 and 1. Logistic Regression is the best calibrated (ECE 0.018). The neural models are over-confident, as Guo et al. (2017) report for modern networks, and AfriBERTa has the highest calibration error of the three (0.049) despite the best log loss. If its probabilities were used to decide when to ask an editor, they would first need recalibrating, for example with temperature scaling (Guo et al., 2017).

5.3 Interpretation

The evidence answers the research question in three parts.

- Word order adds little for topic classification on this data.** A linear model over word and character n-grams matched or beat two neural architectures designed to model order. This agrees with Wang and Manning (2012) and Joulin et al. (2017): the topic of a news article is carried by *which* words occur, such as *timu*, *ligi* or *saratani*, and our EDA shows that these words are distinctive and appear early. The BiLSTM and TextCNN must learn 15 million embedding parameters from 20,000 articles, and the learning curves show them overfitting within a few epochs.
- Pretraining is what moved the hard classes.** AfriBERTa’s gains over Logistic Regression are largest on economy (0.83 vs 0.67) and health (0.62 vs 0.56), the two classes that share vocabulary with national news. Distinguishing a ministry’s budget speech (national) from a tax policy story (economy) needs more than keyword counts, and the contextual representations AfriBERTa learned from unlabelled African-language text (Ogueji et al., 2021) supply it. This matches MasakhaNEWS, where African-centric pretrained models were strongest (Adelani et al., 2023).
- More text and more tricks did not help the from-scratch models.** Because the topic signal is concentrated in the opening sentences (Figure 3), doubling the input to 512 words or tokens did not help any of the three neural models. Class weighting, which transformed Logistic Regression, did not help them, probably because they already reach high recall on the large classes and the weighting mainly increases confident mistakes on the small ones (validation log loss rose from 0.354 to 0.524 for the BiLSTM).

The trade-off is cost. Logistic Regression trains in minutes on a laptop CPU, is the best-calibrated model and is easy to inspect. AfriBERTa needs a GPU and about 42 minutes of fine-tuning per run for

0.051 more macro-F1. For a newsroom tagging tool that suggests topics to an editor, the linear model may be the pragmatic choice; for fully automatic routing of health and economy news, the transformer’s advantage on exactly those classes matters.

6. Error Analysis and Limitations

6.1 Where the models fail

Economy versus national news. The most frequent confusion for the best model is *uchumi* predicted as *kitaiifa* (100 articles, 18% of its errors). Confusion between economy and national news, in both directions, is the largest error type for every model, followed by health versus national (Figure 5). Tanzanian economic news in this corpus is largely about the state: the revenue authority, ministries, public investment programmes. These articles use the same vocabulary as national news (*serikali*, *waziri*, *mradi*), and national news is eight times larger, so models lean towards it. Naive Bayes almost never predicts economy (F1 0.14).

Health. Health is the smallest class and has the lowest F1 for every model except Naive Bayes. In the articles we read, many health stories are framed as government announcements (a minister launching an eye clinic) or as crime (a regional commissioner ordering arrests over stolen government medicines), so their vocabulary overlaps with national news.

Long articles. Every model loses accuracy on the 162 test articles longer than 1,024 words (Figure 9). The drop is largest for the BiLSTM and TextCNN, which read only the first 256 words (0.821 and 0.821), smaller for AfriBERTa, which reads the first 256 subword tokens (0.925 overall, 0.883 on these articles), and smallest for Logistic Regression, which reads the whole article (0.889 overall, 0.883). In a long article the opening is a smaller share of the content, so truncation discards more of the evidence.

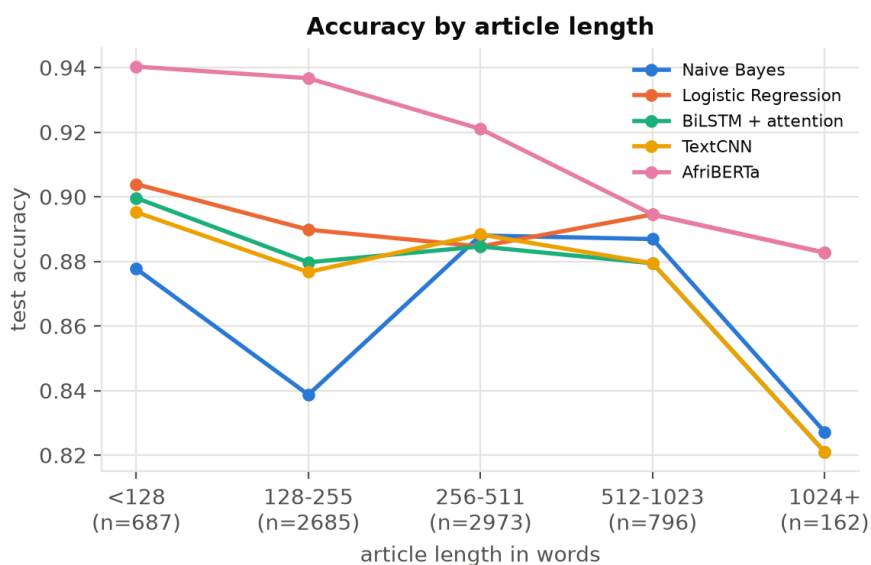


Figure 9. Test accuracy by article length.

Rare words and code-switching. Accuracy barely changes across quartiles of rare-word share for any model (Section 3 showed a 2.5% out-of-vocabulary rate), so vocabulary coverage is not a major source of error. Only 160 test articles have more than 1% English function words. Most models do as well or better on them (Logistic Regression macro-F1 0.84 vs 0.81 on Swahili-only articles), because

code-switching is concentrated in entertainment, an easy class. The BiLSTM is the exception (0.69 vs 0.79), although with only 160 articles this estimate is noisy.

6.2 Are the errors really errors?

5,680 of the 7,303 test articles are classified correctly by all five models, and 243 are misclassified by all five. Reading these articles suggested that many “errors” are labelling problems. To estimate how common this is, we drew a random sample of 30 of the 211 test articles that Logistic Regression misclassified with more than 90% confidence and judged each one by reading its opening (`results/error_label_audit.csv`):

- **17 of 30 (57%) appear mislabelled.** Examples include a report on Azam Federation Cup quarter-final results labelled *national*, a nutrition column labelled *national* (predicted *health*), an Al-Shabaab attack in Kenya labelled *national* (predicted *international*), and a story about a regional commissioner warning cosmetics smugglers labelled *sports*.
- **10 (33%) are genuinely ambiguous**, such as a presidential speech urging pension funds to invest in industry (*economy* or *national*) or a court ruling on the Simba club election (*sports* or *national*).
- **Only 3 (10%) are clear model mistakes.**

The dataset card does not describe how labels were assigned; the pattern of errors suggests they come from the section of the website where each article was published, not from reading the article. This means the reported test scores underestimate how well the models match human judgement, and it sets a ceiling on any model evaluated against these labels.

6.3 Shortcuts

22% of training articles open with a reporter byline (“*Na KULWA MZEE – DAR ES SALAAM*”), and reporters tend to cover one beat. For the 43 reporters with at least 15 training articles, simply predicting each reporter’s most common topic is correct for 87.2% of their 890 test articles. The BiLSTM’s attention suggests the models notice this: in two of the six articles we inspected, a reporter’s name was among the most-attended words, and for one health article the three most-attended words were the reporter’s name and the dateline (`results/error_bilstm_attention.csv`), although attention weights are only a partial explanation of model behaviour (Jain & Wallace, 2019). To test how much the models *depend* on the shortcut, we deleted the names of the 182 reporters seen at least three times and retrained the final Logistic Regression. Macro-F1 fell only slightly (validation 0.832 → 0.825, test 0.817 → 0.815), so the content of the articles, not the byline, carries the prediction. Named entities from one period are a subtler shortcut (Geirhos et al., 2020): *Magufuli*, *Trump* and *Mugabe* are among the most class-associated words, and a model trained on news from the late 2010s may lose accuracy on news from later years.

6.4 Limitations of this study

- **One dataset, one country, one period.** All articles come from a small number of Tanzanian news websites. We cannot claim the results hold for Kenyan Swahili, social media text or news from other years.
- **Noisy labels.** Section 6.2 suggests a substantial share of apparent errors are label errors, so small differences between the strongest models should be read with caution.
- **Limited compute.** AfriBERTa used two seeds instead of three, and class weighting was not tested for it. Larger African models such as AfroXLMR (Alabi et al., 2022) or SwahBERT (Martin et al., 2022) were out of reach.

- **Greedy, single-seed selection.** Each experiment step was judged on one seed, and the seed spread is as large as many step differences, so some rejected changes may be neutral rather than harmful.
- **One validation split.** We used a single stratified hold-out set rather than cross-validation for the neural models, because of training time.

6.5 Responsible AI considerations

A topic classifier looks harmless, but it decides what readers see. The models learned the editorial categories of a few Tanzanian outlets, including their mistakes, and they rely partly on named entities and reporter names, which could disadvantage stories about under-covered regions or by new reporters. Misrouting health news is the highest-stakes failure, and health is also one of the weakest classes. We therefore recommend using such a system to *suggest* topics to an editor rather than to publish automatically, reporting per-class scores rather than accuracy, and re-evaluating on recent news before deployment. Following Bender and Friedman (2018), a data statement for this corpus should document its sources, dates and labelling process, which are currently missing. The data is public under CC BY 4.0 and contains the names of public figures and journalists but no private user data.

7. Conclusion and Future Work

This study asked how effectively sequential models classify Swahili news and what explains their strengths and limitations. On a cleaned, leak-free split of the Swahili News Classification dataset, the fine-tuned AfriBERTa transformer was the best model (0.868 test macro-F1), significantly ahead of a tuned TF-IDF Logistic Regression (0.817). A BiLSTM with attention (0.795) and a TextCNN (0.791), both trained from scratch, did not beat the linear baseline. The evidence points to two explanations: topic is signalled by distinctive words that appear early in an article, so modelling word order adds little, while pretraining on African-language text helps most on the overlapping economy, national and health classes. Error analysis showed that many remaining errors are label errors or genuinely ambiguous articles, and that bylines offer a shortcut the models notice but do not depend on.

Future work should (1) re-annotate a sample of the test set to measure label noise properly and train with noise-robust losses; (2) evaluate larger African-centric models such as AfroXLMR and SwahBERT and parameter-efficient fine-tuning; (3) test on news from other years and from Kenyan and Ugandan outlets to measure temporal and regional drift; and (4) mask named entities during training to reduce dependence on period-specific names.

Use of AI Tools

We used an AI assistant (Claude, by Anthropic) substantially in this project. It drafted most of the code, ran the first round of experiments, proposed the label-audit judgments in Section 6.2 and wrote the first draft of this report. The team chose the challenge, approved the five approaches, reviewed the code, re-ran each member’s experiments on Colab to confirm the results, checked every audit judgment against the article text, and edited the report. All numbers come from recorded runs in `results/experiment_log.csv`. Each member can explain and defend the parts listed for them in the contribution tracker.

References

- Adelani, D. I., Masiak, M., Azime, I. A., Alabi, J. O., Tonja, A. L., Mwase, C., Ogundepo, O., Dossou, B. F. P., Oladipo, A., Nixdorf, D., et al. (2023). MasakhaNEWS: News topic classification for African languages. In *Proceedings of the 13th International Joint Conference on Natural Language Processing and the 3rd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics* (pp. 144–159). Association for Computational Linguistics.
- Alabi, J. O., Adelani, D. I., Mosbach, M., & Klakow, D. (2022). Adapting pre-trained language models to African languages via multilingual adaptive fine-tuning. In *Proceedings of the 29th International Conference on Computational Linguistics* (pp. 4336–4349).
- Bahdanau, D., Cho, K., & Bengio, Y. (2015). Neural machine translation by jointly learning to align and translate. In *International Conference on Learning Representations*.
- Bender, E. M., & Friedman, B. (2018). Data statements for natural language processing: Toward mitigating system bias and enabling better science. *Transactions of the Association for Computational Linguistics*, 6, 587–604.
- Bojanowski, P., Grave, E., Joulin, A., & Mikolov, T. (2017). Enriching word vectors with subword information. *Transactions of the Association for Computational Linguistics*, 5, 135–146.
- Davis, D. (2020). *Swahili: News classification dataset* (Version 0.2) [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.5514203>
- Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT 2019* (pp. 4171–4186).
- Dietterich, T. G. (1998). Approximate statistical tests for comparing supervised classification learning algorithms. *Neural Computation*, 10(7), 1895–1923.
- Dror, R., Baumer, G., Shlomov, S., & Reichart, R. (2018). The hitchhiker’s guide to testing statistical significance in natural language processing. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics* (pp. 1383–1392).
- Geirhos, R., Jacobsen, J.-H., Michaelis, C., Zemel, R., Brendel, W., Bethge, M., & Wichmann, F. A. (2020). Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2(11), 665–673.
- Grave, E., Bojanowski, P., Gupta, P., Joulin, A., & Mikolov, T. (2018). Learning word vectors for 157 languages. In *Proceedings of the Eleventh International Conference on Language Resources and Evaluation (LREC 2018)*.
- Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. (2017). On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning* (pp. 1321–1330).
- Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), 1735–1780.
- Jain, S., & Wallace, B. C. (2019). Attention is not explanation. In *Proceedings of NAACL-HLT 2019* (pp. 3543–3556).
- Joshi, P., Santy, S., Budhiraja, A., Bali, K., & Choudhury, M. (2020). The state and fate of linguistic diversity and inclusion in the NLP world. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* (pp. 6282–6293).
- Joulin, A., Grave, E., Bojanowski, P., & Mikolov, T. (2017). Bag of tricks for efficient text classification. In *Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics* (pp. 427–431).
- Kim, Y. (2014). Convolutional neural networks for sentence classification. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing* (pp. 1746–1751).
- Loshchilov, I., & Hutter, F. (2019). Decoupled weight decay regularization. In *International Conference on Learning Representations*.
- Martin, G. L., Mswahili, M. E., Jeong, Y.-S., & Woo, J. (2022). SwahBERT: Language model of Swahili. In *Proceedings of NAACL-HLT 2022* (pp. 303–313).
- Ogueji, K., Zhu, Y., & Lin, J. (2021). Small data? No problem! Exploring the viability of pretrained multilingual language models for low-resourced languages. In *Proceedings of the 1st Workshop on Multilingual Representation Learning* (pp. 116–126).
- Opitz, J., & Burst, S. (2019). *Macro F1 and macro F1* (arXiv:1911.03347). arXiv.
- Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., et al. (2019). PyTorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems* 32.

- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
- Rennie, J. D. M., Shih, L., Teevan, J., & Karger, D. R. (2003). Tackling the poor assumptions of naive Bayes text classifiers. In *Proceedings of the 20th International Conference on Machine Learning* (pp. 616–623).
- Schuster, M., & Paliwal, K. K. (1997). Bidirectional recurrent neural networks. *IEEE Transactions on Signal Processing*, 45(11), 2673–2681.
- Sokolova, M., & Lapalme, G. (2009). A systematic analysis of performance measures for classification tasks. *Information Processing & Management*, 45(4), 427–437.
- Sun, C., Qiu, X., Xu, Y., & Huang, X. (2019). How to fine-tune BERT for text classification? In *Chinese Computational Linguistics (CCL 2019)* (pp. 194–206). Springer.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. In *Advances in Neural Information Processing Systems* 30.
- Wang, S., & Manning, C. D. (2012). Baselines and bigrams: Simple, good sentiment and topic classification. In *Proceedings of the 50th Annual Meeting of the Association for Computational Linguistics* (pp. 90–94).
- Wolf, T., Debut, L., Sanh, V., Chaumond, J., Delangue, C., Moi, A., et al. (2020). Transformers: State-of-the-art natural language processing. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations* (pp. 38–45).
- Yang, Z., Yang, D., Dyer, C., He, X., Smola, A., & Hovy, E. (2016). Hierarchical attention networks for document classification. In *Proceedings of NAACL-HLT 2016* (pp. 1480–1489).
- Zhang, X., Zhao, J., & LeCun, Y. (2015). Character-level convolutional networks for text classification. In *Advances in Neural Information Processing Systems* 28.
- Zindi. (2021). *AI4D Swahili news classification challenge*. <https://zindi.africa/competitions/ai4d-swahili-news-classification-challenge>

Appendix A. Experiment log

Table A1. Experiment log (validation macro-F1; test macro-F1 only for selected configurations). Grid-search rows BO6 and LO4 are summarised by the selected row.

ID	Model	Change	Val macro-F1	Test macro-F1
B01	Naive Bayes	Raw lowercased text, word unigrams	0.691	
B02	Naive Bayes	Cleaning: split glued sentences, repair ligatures, drop punctuation and numbers	0.690	
B03	Naive Bayes	Cleaning + Swahili stopword removal	0.697	
B04	Naive Bayes	Word unigrams + bigrams	0.684	
B05	Naive Bayes	ComplementNB (designed for imbalanced text), unigrams + bigrams	0.657	
B07	Naive Bayes	Final: ComplementNB alpha=0.03, cleaned text, unigrams + bigrams	0.700	0.689
C01	TextCNN	TextCNN, random embeddings, kernels 3/4/5, first 256 words	0.808	
C02	TextCNN	Read the first 512 words instead of 256	0.787	
C03	TextCNN	Initialise with fastText Swahili vectors	0.801	
C04	TextCNN	Class-weighted cross-entropy	0.807	
C05	TextCNN	200 filters and kernel sizes 2 to 5	0.770	
C06	TextCNN	Final configuration, 3 seeds (test macro-F1 0.7813 +/- 0.0093)	0.808	0.791
L01	Logistic Regression	Cleaned text, word unigrams + bigrams, no class weights	0.791	
L02	Logistic Regression	Balanced class weights	0.832	
L03	Logistic Regression	Add character 2-5 grams	0.832	
L05	Logistic Regression	Final: C=10, balanced weights, word 1-2 grams + char 2-5 grams	0.832	0.817
R01	BiLSTM + attention	BiLSTM, random embeddings, final hidden state, first 256 words	0.797	
R02	BiLSTM + attention	Attention pooling instead of the final hidden state	0.804	
R03	BiLSTM + attention	Read the first 512 words instead of 256	0.798	
R04	BiLSTM + attention	Initialise with fastText Swahili vectors	0.794	
R05	BiLSTM + attention	Class-weighted cross-entropy	0.792	
R06	BiLSTM + attention		0.804	0.795

ID	Model	Change	Val macro-F1	Test macro-F1
		Final configuration, 3 seeds (test macro-F1 0.7998 +/- 0.0043)		
T01	AfriBERTa	AfriBERTa-small, first 256 subword tokens, lr 5e-5, 3 epochs	0.880	
T02	AfriBERTa	First 512 tokens instead of 256	0.868	
T03	AfriBERTa	Keep the first 128 and last 382 tokens	0.875	
T04	AfriBERTa	Final configuration, 2 seeds (test macro-F1 0.8697 +/- 0.0014)	0.880	0.868